

Daily Geometry Drill #3 — Answers & Investigations

Sheet by Henry Koduthore

Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

New toolkit today: Relative Positions · Radical Axis by subtraction · Common Chords · Orthogonal Circles · Tangency to Both Axes.

Section A Rapid Recognition

1. The circle $x^2 + y^2 - 12x - 16y + 75 = 0$ is tangent to the circle $x^2 + y^2 = 25$. Write down the number of common tangents of the two circles.

Answer: 3

Method: $x^2 + y^2 - 12x - 16y + 75 = (x - 6)^2 + (y - 8)^2 - 100 + 75 = 0$, so centre (6, 8), radius $\sqrt{36 + 64 - 75} = 5$. Both circles have radius 5; $d = 10 = r_1 + r_2$ — externally tangent — 3 common tangents.

Investigate further: The bookkeeping: $d > r_1 + r_2 \Rightarrow 4$, $d = r_1 + r_2 \Rightarrow 3$, $|r_1 - r_2| < d < r_1 + r_2 \Rightarrow 2$, $d = |r_1 - r_2| \Rightarrow 1$, $d < |r_1 - r_2| \Rightarrow 0$. **“Tangent” means exactly one common point, giving exactly 3 tangents, not 2.**

2. The circles with centre (0, 0) radius 5 and centre (13, 0) radius 3 are disjoint. Write down the number of common tangents of the two circles.

Answer: 4

Method: $d = 13$, $r_1 + r_2 = 8$; since $d > r_1 + r_2$ the circles are fully disjoint and there are 4 common tangents.

Investigate further: Disjoint circles admit 2 direct (external) tangents plus 2 transverse (internal) tangents. **Once the circles overlap, the two transverse tangents disappear.**

3. Write down the radical axis of the circles $x^2 + y^2 - 6x + 2y - 11 = 0$ and $x^2 + y^2 - 2x - 4y + 1 = 0$.

Answer: $2x - 3y + 6 = 0$

Method: At the radical axis the powers are equal, so subtract: $(-6x + 2y - 11) - (-2x - 4y + 1) = 0$, i.e. $-4x + 6y - 12 = 0$, i.e. $2x - 3y + 6 = 0$. The $x^2 + y^2$ terms cancel in one stroke.

Investigate further: Check: (0, 2) lies on $2x - 3y + 6 = 0$, and its power is $4 + 4 - 11 = -3$ for the first circle and $4 - 8 + 1 = -3$ for the second. Before subtracting, check both $r^2 > 0$ (21 and 4 here): subtraction gives a line even when one “circle” has no real points.

4. Write down whether the circles $x^2 + y^2 = 9$ and $x^2 + y^2 + 6x + 8y + 9 = 0$ are orthogonal (yes or no).

Answer: yes

Method: $x^2 + y^2 + 6x + 8y + 9$ has centre $(-3, -4)$ and $r^2 = 9 + 16 - 9 = 16$, so $r = 4$. $d = \sqrt{3^2 + 4^2} = 5$. Orthogonal iff $d^2 = r_1^2 + r_2^2$: $25 = 9 + 16$ ✓. Yes.

Investigate further: Equivalently with coefficients: $g_1 = f_1 = 0$, $g_2 = 3$, $f_2 = 4$ and $2g_1g_2 + 2f_1f_2 = 0$ must equal $c_1 + c_2 = (-9) + (9) = 0$. **Orthogonality is about the radii forming a right triangle with the centre distance.**

5. The circles $x^2 + y^2 = 25$ and $x^2 + y^2 - 10x + 5 = 0$ intersect in two points. Find the length of their common chord.

Answer: 8

Method: Subtract: $-25 = -10x + 5 \Rightarrow x = 3$, the radical axis. Distance of $x = 3$ from the origin is 3, radius 5, so half-chord $= \sqrt{25 - 9} = 4$, chord $= 8$.

Investigate further: The second circle has centre $(5, 0)$ and radius $\sqrt{20}$; the two circles agree on the two end-points of this chord. **Identify the radical axis first — the chord sits on it — then apply the half-chord formula.**

6. Two circles have radii 3 and 5 and their centres are 4 units apart. Write down the number of common tangents.

Answer: 2

Method: $|r_1 - r_2| = 2 < d = 4 < r_1 + r_2 = 8$ — intersecting circles — 2 common tangents.

Investigate further: Two crossing circles always have exactly 2 common tangents. **The trap is counting the two intersection points as “tangents”.**

7. The circle $x^2 + y^2 - 10x - 10y + k = 0$ is tangent to both coordinate axes. Find k .

Answer: 25

Method: Centre $(5, 5)$. Tangent to both axes forces the radius to equal each coordinate: $r = 5$. Then $r^2 = 25 = 5^2 + 5^2 - k$, so $k = 50 - 25 = 25$.

Investigate further: A circle tangent to both axes has centre (r, r) , $(-r, r)$, $(r, -r)$ or $(-r, -r)$ and equation $(x \mp r)^2 + (y \mp r)^2 = r^2$, constant r^2 . **The constant is r^2 , not $2r^2$.**

8. The circle $x^2 + y^2 + 4x - 2y + C = 0$ is orthogonal to the circle $x^2 + y^2 - 4x + 4y - 1 = 0$. Find C .

Answer: -11

Method: $g_1 = 2$, $f_1 = -1$; $g_2 = -2$, $f_2 = 2$, $c_2 = -1$. Orthogonality: $2g_1g_2 + 2f_1f_2 = c_1 + c_2 \Rightarrow 2(2)(-2) + 2(-1)(2) = C - 1 \Rightarrow -8 - 4 = C - 1 \Rightarrow C = -11$.

Investigate further: Geometric check: the centres are $(-2, 1)$ and $(2, -2)$, so $d^2 = 25$; $r_1^2 = 5 - C$ and $r_2^2 = 9$, and $25 = (5 - C) + 9$ gives $C = -11$ ✓. **Both forms of the test say the same thing.**

9. How many common tangents do the circles $x^2 + y^2 - 4x - 6y + 12 = 0$ and $x^2 + y^2 + 2x - 10y + 25 = 0$ have?

Answer: 4

Method: First: centre $(2, 3)$, $r^2 = 4 + 9 - 12 = 1$. Second: centre $(-1, 5)$, $r^2 = 1 + 25 - 25 = 1$. $d = \sqrt{(3)^2 + (2)^2} = \sqrt{13} > 2 = r_1 + r_2$ — disjoint unit circles — 4 common tangents.

Investigate further: Completing the square (the Worksheet #2 skill) locates the centres; only then compare d with $r_1 + r_2$. **Two unit circles need $d > 2$ for four tangents; here $d = \sqrt{13}$.**

10. Two circles have radii 7 and 3 and exactly 3 common tangents. Find the distance between their centres.

Answer: 10

Method: Exactly 3 common tangents means the circles are externally tangent, so $d = r_1 + r_2 = 7 + 3 = 10$.

Investigate further: Keep the five-case table on one line. **If there were 2 tangents you would instead need $|3 - 7| < d < 10$, which leaves a whole range, not a single value.**

Section B Manipulation Drills

1. How many common tangents do the circles $x^2 + y^2 - 2x - 4y - 20 = 0$ and $x^2 + y^2 - 6x - 2y + 6 = 0$ have?

- A) 0
B) 1
C) 2
D) 3
E) 4

Answer: A) 0

Method: First circle: centre $(1, 2)$, $r_1^2 = 1 + 4 + 20 = 25$, so $r_1 = 5$. Second: centre $(3, 1)$, $r_2^2 = 9 + 1 - 6 = 4$, so $r_2 = 2$. Then $d = \sqrt{4 + 1} = \sqrt{5} < 3 = r_1 - r_2$, so the small circle lies strictly inside the large one and there are no common tangents. Option C is what you get by checking only $d < r_1 + r_2$.

Investigate further: Slide the small circle directly away from $(1, 2)$. At what centre distance does the count first become 1, and what do the circles look like there?

2. The circle $x^2 + y^2 + ax + by + c = 0$ passes through the origin, and its radical axis with the circle $x^2 + y^2 + 6x - 4y + 10 = 0$ is the line $2x - y + 5 = 0$. What is $a + b$?

- A) 3
B) 5
C) -3
D) 0

Answer: D) 0

Method: Through the origin means $c = 0$. Subtracting gives $(a - 6)x + (b + 4)y - 10 = 0$, and this must be a multiple $k(2x - y + 5)$ of the given line. The constants give $5k = -10$, so $k = -2$. Then $a - 6 = -4$ and $b + 4 = 2$, so $a = 2$, $b = -2$ and $a + b = 0$. Option A comes from matching coefficients with $k = 1$, but the equation of a line is only fixed up to a multiple.

Investigate further: Drop the condition $c = 0$. Show that $a + b$ can then take any value, and say what that tells you about how much a radical axis determines one of the circles.

3. The circle $x^2 + y^2 - 4x + 2y - 5 = 0$ is orthogonal to $x^2 + y^2 + 6x - 2y + k = 0$. Find k .

- A) -9
- B) 9
- C) -5
- D) 14

Answer: A) -9

Method: $g_1 = -2, f_1 = 1, c_1 = -5; g_2 = 3, f_2 = -1, c_2 = k$. $2g_1g_2 + 2f_1f_2 = 2(-2)(3) + 2(1)(-1) = -14 = c_1 + c_2 = -5 + k \Rightarrow k = -9$.

Investigate further: Orthogonal circles cut at right angles: the radii to a common point are perpendicular. **Never pair g with c or f with g — the pairing is $g_1g_2 + f_1f_2$ against $\frac{c_1+c_2}{2}$.**

4. A circle is tangent to both coordinate axes in the first quadrant and passes through the point $(1, 8)$. Find the smaller possible radius.

Answer: 5

Method: Centre (r, r) , radius r : $(1 - r)^2 + (8 - r)^2 = r^2 \Rightarrow r^2 - 18r + 65 = 0 \Rightarrow (r - 5)(r - 13) = 0$. Smaller radius 5.

Investigate further: There really are two such circles — always state which root the question wants. **Both roots are positive, so the phrasing “smaller” is essential.**

5. Two circles of equal radius 5 have centres $(0, 0)$ and $(6, 0)$. Their radical axis has equation?

- A) $x = 3$
- B) $x = 6$
- C) $y = 3$
- D) $y = 0$

Answer: A) $x = 3$

Method: Equal radii: subtract $(x^2 + y^2 - 25) = ((x - 6)^2 + y^2 - 25) \Rightarrow -25 = -12x + 36 - 25 \Rightarrow x = 3$.

Investigate further: For equal circles the radical axis is the perpendicular bisector of the centre segment. **A quick sanity check: power at $(3, 0)$ is $9 - 25 = -16$ for both sides.**

6. Two circles of radii 4 and 6 have centres 9 units apart. How many common tangents do they have?

- A) 0
- B) 1
- C) 2
- D) 3
- E) 4

Answer: C) 2

Method: $|r_1 - r_2| = 2 < d = 9 < 10 = r_1 + r_2$: strict overlap without containment — 2 common tangents.

Investigate further: Compare d to 10 (sum) and 2 (difference) before counting. 9 is **dangerously close to 10; only exactly 10 would give 3**.

7. A circle with centre $(7, 1)$ cuts the circle $x^2 + y^2 - 2x + 4y - 4 = 0$ at right angles. What is its radius?
- A) $3\sqrt{5}$
 B) $3\sqrt{5} - 3$
 C) 3
 D) 6

Answer: D) 6

Method: At a common point the two radii are perpendicular, so the new radius is a tangent to the given circle: R is the tangent length from $(7, 1)$, and R^2 is the power of $(7, 1)$, namely $49 + 1 - 14 + 4 - 4 = 36$. So $R = 6$. Check: the given circle has centre $(1, -2)$ and $r = 3$, and $d^2 = 36 + 9 = 45 = R^2 + r^2$. Option A is d ; option B is $d - r$, the radius of the circle that just touches.

Investigate further: Show that every circle centred on the line $y = 1$ and orthogonal to the given circle passes through $(1, 1)$. What is special about the line $y = 1$?

8. The circles $x^2 + y^2 = 25$ and $x^2 + y^2 - 12x - 6y + 25 = 0$ meet at P and Q . Find the midpoint of PQ .

Answer: $(\frac{10}{3}, \frac{5}{3})$

Method: The common chord lies on the radical axis: subtracting gives $12x + 6y - 50 = 0$, i.e. $6x + 3y = 25$. The midpoint of PQ is where this meets the line of centres, which joins $(0, 0)$ to $(6, 3)$, so $y = \frac{x}{2}$. Then $6x + \frac{3}{2}x = 25$, so $x = \frac{10}{3}$ and $y = \frac{5}{3}$. The midpoint of the centres, $(3, \frac{3}{2})$, is wrong because the radii (5 and $2\sqrt{5}$) differ.

Investigate further: Find PQ from the midpoint M alone, without finding P or Q . (Use OM and the radius 5; you should get $\frac{20}{3}$.)

9. A circle with its centre in the first quadrant touches both coordinate axes and the line $3x + 4y = 12$. What is the sum of the possible radii?
- A) 1
 B) 6
 C) 7
 D) 5

Answer: C) 7

Method: The centre is (r, r) , and its distance from the line is $r: \frac{|7r-12|}{5} = r$, so $7r - 12 = \pm 5r$, giving $r = 6$ or $r = 1$. Both circles exist: $r = 1$ is the incircle of the triangle with vertices $(0, 0)$, $(4, 0)$, $(0, 3)$, and $r = 6$ lies beyond the line and touches it from the far side. The sum is 7. Option A is the incircle alone, from dropping the modulus.

Investigate further: Allow the centre in any quadrant. How many circles touch both axes and the line, and why are they exactly the incircle and the three excircles of the 3–4–5 triangle?

10. The point $P = (1, t)$ lies on the radical axis $3x - 4y + 5 = 0$ of two circles. Find t .

Answer: 2

Method: Substitute: $3(1) - 4t + 5 = 0 \Rightarrow 8 - 4t = 0 \Rightarrow t = 2$.

Investigate further: The radical axis is exactly the set of points of equal power — a line. **Put P into the line, not into either circle equation.**

Section C Substitution & Structure

1. Two circles have the same radius. Their centres are $C_1(-2, 1)$ and $C_2(3, -2)$. The circles intersect in two distinct points P and Q . Which is the equation of the line PQ ? (after TMUA 2021 Paper 1 Q1)

- A) $5x + 3y = 4$
 B) $3x - 5y = 4$
 C) $5x - 3y = 4$
 D) $5x - 3y = 1$

Answer: C) $5x - 3y = 4$

Method: Equal radii: subtract $(x+2)^2 + (y-1)^2 = (x-3)^2 + (y+2)^2$, i.e. $x^2 + 4x + 4 + y^2 - 2y + 1 = x^2 - 6x + 9 + y^2 + 4y + 4$, so $10x - 6y = 8$, i.e. $5x - 3y = 4$. The $x^2 + y^2$ cancel in one line. The midpoint $(\frac{1}{2}, -\frac{1}{2})$ satisfies $5(\frac{1}{2}) - 3(-\frac{1}{2}) = 4$ ✓.

Investigate further: Equal radii make the radical axis the perpendicular bisector of C_1C_2 , but you still get it fastest by subtraction. **Option A flips the sign of y ; option D is parallel to the right line but misses the midpoint $(\frac{1}{2}, -\frac{1}{2})$.**

2. The circles $C_1 : (x-1)^2 + y^2 = 4$ and $C_2 : (x-5)^2 + (y-3)^2 = 9$ have radii 2 and 3. How many common tangents do C_1 and C_2 have? (after TMUA 2019 Paper 1 Q7)

- A) 0
 B) 1
 C) 2
 D) 3
 E) 4

Answer: D) 3

Method: $C_1 = (1, 0)$, $C_2 = (5, 3)$, $d = \sqrt{4^2 + 3^2} = 5$. $r_1 + r_2 = 5$, so $d = r_1 + r_2$ — externally tangent — exactly 3 common tangents (2 direct plus the common tangent at the touch point).

Investigate further: $d > r_1 + r_2 \Rightarrow 4$, $d = r_1 + r_2 \Rightarrow 3$, $|r_1 - r_2| < d < r_1 + r_2 \Rightarrow 2$, $d = |r_1 - r_2| \Rightarrow 1$, $d < |r_1 - r_2| \Rightarrow 0$. **The 3-4-5 triple makes d instant.**

3. The circles $x^2 + y^2 = 169$ and $x^2 + y^2 - 6x + 8y - 119 = 0$ meet at P and Q . What is the length PQ ?

- A) 12
 B) 16

C) 24

D) 26

Answer: C) 24

Method: Second circle: centre $(3, -4)$, $r^2 = 9 + 16 + 119 = 144$, so $r = 12$. Subtracting gives the radical axis $6x - 8y - 50 = 0$, i.e. $3x - 4y = 25$. It is $\frac{25}{5} = 5$ from the origin, so the half-chord in the big circle is $\sqrt{169 - 25} = 12$ and $PQ = 24$. Check: $(3, -4)$ satisfies $3x - 4y = 25$, so the chord passes through the second centre and is a diameter of the smaller circle, $2 \times 12 = 24$. Option A is the half-chord; option D is the diameter of the larger circle.

Investigate further: The chord runs through the smaller circle's centre because $d^2 + r_2^2 = r_1^2$ ($25 + 144 = 169$). Explain why that condition forces it, using the right angle between the chord and the line of centres.

4. A circle in the first quadrant is tangent to both coordinate axes and passes through the point $(9, 2)$. Its two possible radii are $r_1 < r_2$. What is $r_1 + r_2$? (*adapted from JZMaths Mock 1 Q5*)

A) 14

B) 17

C) 22

D) 34

Answer: C) 22

Method: Centre (r, r) , radius r : $(9 - r)^2 + (2 - r)^2 = r^2 \Rightarrow r^2 - 22r + 85 = 0$. By Vieta sum = 22, product = 85; roots are 5 and 17. Sum 22.

Investigate further: Both $r = 5$ and $r = 17$ give genuine circles. **Option B 17 is the larger root alone; option A 14 would be $9 + 5$ misreading.**

5. The circles $x^2 + y^2 = 25$ and $x^2 + y^2 - 16x - 8y - 1 = 0$ meet at P and Q . What is the radius of the circle through P , Q and the point $(0, 1)$?

A) 9

B) $2\sqrt{13}$ C) $4\sqrt{5}$

D) 13

Answer: D) 13

Method: Every circle through P and Q has the form $S_1 + \lambda S_2 = 0$, where $S_1 = x^2 + y^2 - 25$ and $S_2 = x^2 + y^2 - 16x - 8y - 1$. At $(0, 1)$, $S_1 = -24$ and $S_2 = -8$, so $\lambda = -3$. Then $S_1 - 3S_2 = -2x^2 - 2y^2 + 48x + 24y - 22$, which gives $x^2 + y^2 - 24x - 12y + 11 = 0$: centre $(12, 6)$, $r^2 = 144 + 36 - 11 = 169$, $r = 13$. Option B comes from $\lambda = +3$; option A is the radius of the second circle; option C is the distance between the centres.

Investigate further: P and Q have irrational coordinates here, and the family never needs them. Which member of the family $S_1 + \lambda S_2 = 0$ is a straight line, and which line is it?

6. A circle has equation $x^2 + ax + y^2 + by + c = 0$ where $a, b, c \neq 0$. Which condition is necessary

and sufficient for the circle to be tangent to the y -axis? (after TMUA 2021 Paper 2 Q15)

- A) $a^2 = 4c$
- B) $b^2 = 4c$
- C) $a^2 + b^2 = 4c$
- D) $c = 0$

Answer: B) $b^2 = 4c$

Method: Centre $(-a/2, -b/2)$, $r^2 = a^2/4 + b^2/4 - c$. Tangent to y -axis ($x = 0$) needs distance $|-a/2| = r$, so $a^2/4 = a^2/4 + b^2/4 - c \Rightarrow b^2 = 4c$. (Tangency to x -axis would be $a^2 = 4c$.)

Investigate further: The y -axis condition kills b , not a , because the radius condition cancels $a^2/4$.
Option A $a^2 = 4c$ is tangency to the x -axis; **option D** $c = 0$ is through the origin.

7. The equal circles $C_1 : (x + 2)^2 + (y - 1)^2 = 3$ and $C_2 : (x - 4)^2 + (y - 1)^2 = 3$ have centres 6 apart. A transverse common tangent with positive gradient makes an acute angle θ with the x -axis. What is $\tan \theta$? (adapted from TMUA 2020 Paper 1 Q16)

- A) $\frac{1}{2}$
- B) $\frac{\sqrt{2}}{2}$
- C) $\sqrt{2}$
- D) $\frac{\sqrt{3}}{3}$

Answer: B) $\frac{\sqrt{2}}{2}$

Method: Equal radii $\sqrt{3}$ at $(-2, 1)$ and $(4, 1)$. Transverse tangent passes through midpoint $(1, 1)$: line $y - 1 = m(x - 1)$, i.e. $mx - y + 1 - m = 0$. Distance from $(-2, 1)$ to line $= |-3m|/\sqrt{m^2 + 1} = \sqrt{3} \Rightarrow 9m^2 = 3(m^2 + 1) \Rightarrow 6m^2 = 3 \Rightarrow m^2 = 1/2 \Rightarrow m = \sqrt{2}/2$ (positive).

Investigate further: The transverse tangent crosses the centre line at the midpoint; the direct common tangents are horizontal ($m = 0$). **Option C** $\sqrt{2}$ inverts the fraction; **option A** $1/2$ squares the root.

8. The three circles $C_1 : x^2 + y^2 + 4x - 1 = 0$, $C_2 : x^2 + y^2 - 8y + 11 = 0$, $C_3 : x^2 + y^2 + 2x - 12y + 13 = 0$ have pairwise radical axes concurrent at R . What are the coordinates of R ? (adapted from Vantage Mock Paper 2 Q10 / BMO1 2017 Q4)

- A) $(1, 1)$
- B) $(1, -1)$
- C) $(3, 1)$
- D) $(1, 2)$

Answer: A) $(1, 1)$

Method: Radical axis $C_1 - C_2$: $(4x - 1) - (-8y + 11) = 0 \Rightarrow 4x + 8y - 12 = 0 \Rightarrow x + 2y - 3 = 0$.
 $C_2 - C_3$: $(-8y + 11) - (2x - 12y + 13) = 0 \Rightarrow -2x + 4y - 2 = 0 \Rightarrow x - 2y + 1 = 0$. Solve $x + 2y = 3$,
 $x - 2y = -1 \Rightarrow x = 1, y = 1$. Power: $1 + 1 + 4 - 1 = 5$ for all three ✓.

Investigate further: Any two radical axes intersect at the radical centre; the third must pass through it. **Option D** $(1, 2)$ lies on none of the three radical axes; substituting rejects it in seconds.

Section D Challenge Ramp

1. The circles $x^2 + y^2 - 4x - 2y - 4 = 0$ and $x^2 + y^2 - 20x - 14y + 124 = 0$ have centres O_1 and O_2 . A common external tangent touches them at P and Q respectively. What is the area of the quadrilateral O_1PQO_2 ?

- A) 24
 B) $16\sqrt{6}$
 C) 40
 D) $8\sqrt{6}$

Answer: B) $16\sqrt{6}$

Method: $O_1 = (2, 1)$ with $r_1 = \sqrt{4 + 1 + 4} = 3$; $O_2 = (10, 7)$ with $r_2 = \sqrt{100 + 49 - 124} = 5$; $d = 10$. Both radii are perpendicular to PQ , so O_1PQO_2 is a right trapezium with parallel sides 3 and 5 and height PQ . Dropping a perpendicular from O_1 to O_2Q gives $PQ = \sqrt{d^2 - (r_2 - r_1)^2} = \sqrt{96} = 4\sqrt{6}$, so the area is $\frac{1}{2}(3 + 5) \cdot 4\sqrt{6} = 16\sqrt{6}$. Option A uses the internal tangent length $\sqrt{d^2 - (r_1 + r_2)^2} = 6$; option C uses d as the height.

Investigate further: The two external tangents meet at a point T on the line O_1O_2 . Use similar triangles to show $TO_1 = 15$, and find the coordinates of T .

2. The circles $(x + 4)^2 + (y + 1)^2 = 64$ and $(x - 8)^2 + (y - 4)^2 = r^2$ ($r > 0$) have exactly one point in common. What is the difference between the two possible values of r ? (after TMUA 2019 Paper 1 Q6)

- A) 8
 B) 16
 C) 26
 D) 42

Answer: B) 16

Method: Centre distance $d = \sqrt{12^2 + 5^2} = 13$, $r_1 = 8$. One point $\Rightarrow |r - 8| = 13$ (internal) or $r + 8 = 13$ (external) $\Rightarrow r = 21$ or 5 , difference 16.

Investigate further: External tangency gives smaller $r = 5$, internal gives larger 21. **Option A** 8 is r_1 itself; **option C** 26 is the sum of the two values, not the difference.

3. Three circles of radii 3, 4 and 5 are pairwise externally tangent. What is the area of the triangle formed by joining their centres? (adapted from BMO1 2017 Q4)

- A) $12\sqrt{5}$
 B) 12

C) 84

D) 30

Answer: A) $12\sqrt{5}$

Method: Centre triangle sides $3 + 4 = 7$, $3 + 5 = 8$, $4 + 5 = 9$. $s = 12$, Heron area $= \sqrt{12 \cdot 5 \cdot 4 \cdot 3} = \sqrt{720} = 12\sqrt{5}$. Coordinate check: $(0, 0)$, $(7, 0)$, third $x = 33/7$, $y = 24\sqrt{5}/7$, area $= \frac{1}{2} \cdot 7 \cdot 24\sqrt{5}/7 = 12\sqrt{5}$ ✓.

Investigate further: Side is sum of radii, not difference. **Option C 84 is area of 13-14-15 triangle, the classic Heron triple.**

4. Two circles of radii 3 and 4 are orthogonal. What is the length of their common chord?

A) $\frac{12}{5}$ B) $\frac{24}{5}$

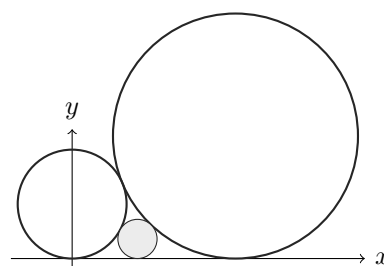
C) 6

D) $\frac{48}{5}$ **Answer:** B) $\frac{24}{5}$

Method: $d^2 = r_1^2 + r_2^2 = 9 + 16 = 25$, so $d = 5$. Place the centres at $(0, 0)$ and $(5, 0)$; the radical axis is $10x = 18$, i.e. $x = \frac{9}{5}$. Half-chord $= \sqrt{9 - \frac{81}{25}} = \frac{12}{5}$, so the chord is $\frac{24}{5}$. Option A is the half-chord; option C is the diameter of the small circle.

Investigate further: The radii to a common point and the line of centres form a right triangle with legs r_1 , r_2 , and the half-chord is its altitude. Deduce that the chord is $\frac{2r_1r_2}{\sqrt{r_1^2 + r_2^2}}$ for any pair of orthogonal circles.

5. The circles $x^2 + y^2 - 8y = 0$ and $x^2 + y^2 - 24x - 18y + 144 = 0$ both touch the x -axis and touch each other. A third circle lies in the gap between them and the x -axis, and touches both circles and the x -axis. What is its radius?

A) $\frac{6}{5}$ B) $\frac{36}{13}$ C) $\frac{169}{100}$ D) $\frac{36}{25}$ **Answer:** D) $\frac{36}{25}$

Method: The circles have centres $(0, 4)$ and $(12, 9)$ and radii 4 and 9, and $d = \sqrt{144 + 25} = 13 = 4 + 9$, so they touch. Key fact: if circles of radii a and b sit on a line and touch each other externally, their points of contact with the line are $\sqrt{(a+b)^2 - (a-b)^2} = 2\sqrt{ab}$ apart. The big pair touches the axis at $x = 0$ and $x = 12 = 2\sqrt{36}$, which checks. The small circle, radius r , touches the axis between them, so

$2\sqrt{4r} + 2\sqrt{9r} = 12$, i.e. $10\sqrt{r} = 12$, $\sqrt{r} = \frac{6}{5}$ and $r = \frac{36}{25}$. Option A forgets to square; option C uses the centre distance 13 in place of the contact distance 12; option B is the guess $\frac{1}{r} = \frac{1}{4} + \frac{1}{9}$.

Investigate further: The same fact gives $\frac{1}{\sqrt{r}} = \frac{1}{\sqrt{a}} + \frac{1}{\sqrt{b}}$ for the circle in the gap. Another circle touches both circles externally and the x -axis but lies outside the gap. Find its radius.

Top 5 Patterns — Worksheet #3

- 1. Classify two circles by comparing d with $r_1 + r_2$ and $|r_1 - r_2|$.** $d > r_1 + r_2$ gives 4 common tangents, $d = r_1 + r_2$ gives 3, overlap 2, internal tangency 1, containment 0. Circles that touch externally have $d = r_1 + r_2$, which also sets the sides of a triangle of centres. (see A1, A2, A6, A9, A10, B1, B6, C2, D2, D3)
- 2. Radical axis = the two equations subtracted.** The $x^2 + y^2$ terms vanish. For intersecting circles it carries the common chord, whose midpoint is where it crosses the line of centres. (see A3, A5, B5, B8, B10, C1, C3, C8)
- 3. Orthogonality three ways.** $d^2 = r_1^2 + r_2^2$; or $2g_1g_2 + 2f_1f_2 = c_1 + c_2$; or the radius of one circle equals the tangent length from its centre to the other, so R^2 is a power. (see A4, A8, B3, B7, D4)
- 4. An equation is only fixed up to a factor.** A radical axis gives $k(ax + by + c)$, and the circles through two intersection points are $S_1 + \lambda S_2 = 0$; one extra condition fixes k or λ . (see B2, C5)
- 5. Tangency means centre-to-line distance = r .** Touching both axes puts the centre at $(\pm r, \pm r)$; a common external tangent has length $\sqrt{d^2 - (r_1 - r_2)^2}$; two touching circles on a line meet it $2\sqrt{ab}$ apart. (see A7, B4, B9, C4, C6, C7, D1, D5)

Common Traps — Worksheet #3

- 1. Testing only $d < r_1 + r_2$.** That does not separate crossing circles from one circle inside the other, and the boundary cases $d = r_1 + r_2$ and $d = |r_1 - r_2|$ each change the count by one. (see A1, A6, A10, B1, B6, C2, D2)
- 2. Matching coefficients as if a line's equation were unique.** $2x - y + 5 = 0$ and $-4x + 2y - 10 = 0$ are the same line; find the scale factor from a known coefficient first. Likewise the sign of λ in a family decides which circle you get. (see B2, C5)
- 3. Using the midpoint of the centres as the midpoint of the chord.** That works only for equal radii; otherwise intersect the radical axis with the line of centres. (see B5, B8, C1)
- 4. Dropping a modulus or a root.** Distance conditions like $\frac{|7r-12|}{5} = r$ and tangency conditions like $|r - 8| = 13$ have two solutions, and the question may want both. (see B4, B9, C4, D2)
- 5. Using the centre distance where a different length is needed.** The tangent length is $\sqrt{d^2 - (r_1 - r_2)^2}$, not d ; the contact points of touching circles on a line are $2\sqrt{ab}$ apart, not $a + b$; and a half-chord is not the chord. (see C3, D1, D4, D5)

“Two circles are solved by one subtraction: locate, subtract, then compare with the sum and difference of the radii.”

Found a mistake or want to contribute a sheet?

Visit the open-source project at github.com/The-CerealDev/speed-maths